R107/A_edge_factorized | Explicit paths and actual latency

Complete standalone model including graph/path construction; no equal-FLOPs claim.

Problem	Instances	Milliseconds / instance
TSP	128	0.937643
CVRP	128	1.127595
ATSP	128	0.108179
OVRP	128	0.128660
VRPB	128	0.128463
VRPL	128	0.128464
VRPTW	128	0.128525
OVRPTW	128	0.128542
OVRPB	128	0.128528
OVRPL	128	0.128326
VRPBL	128	0.128218
VRPBTW	128	0.128335
VRPLTW	128	0.128290
OVRPBL	128	0.128595
OVRPBTW	128	0.128389
OVRPLTW	128	0.128383
VRPBLTW	128	0.128295
OVRPBLTW	128	0.128508

Total trainable parameters: 1,892,233; factorized edge/path evidence

Optional R94B: not_requested

R107 adds an explicit short-path module to the fresh R84/R88B full18 selector. A_edge_factorized uses factorized evidence; B_connected_paths uses joint connected-path computation. This is not a MoE FFN or learned expert-routing experiment. Same complete fresh tensors and total parameter counts require actual factory/initialization proof; they do not imply equal active functional freedom or equal FLOPs. Actual counts and path specification come from the locked model, never estimates. Factorization is conditional on each round incoming node states, not global selector additivity. A retains trunk interactions; B also receives explicit composed fields, so this is not an isolated topology-only causal comparison. Original R88B physical TSP node0 role, methodID+34, geometry/distance, score MLP/maximin and original CE/comparison/risk/two-forward RDropObjective remain; 57,600 updates, logical batch1024. No donor, teacher, context labels, new targets, calibration, quota or auxiliary loss. R89A AdamW, R89B Muon and R84A are saved historical references, not rerun or newly parameter/FLOP-matched controls; R94B is optional, never a prerequisite. Fixed36 rescues, losses, false picks and aligned net costs/regret are separate from choice frequency. Native/FP64 ind and legal pools; TRAIN-fitted SBS; ALL task-equal. TEST is development-exposed; one whole-model full18 VAL checkpoint for full TRAIN/VAL/TEST/LIB; no TEST/LIB reselection. Actual standalone GPU-resident-input timing includes complete model graph/path construction and computation, excludes dataset I/O, CPU transfer and solvers, and is not end-to-end dataset latency. No routing-summary requirement. No online W&B equality/availability is asserted.

R107/A_edge_factorized | Same selected model: full TRAIN

Diagnostic only; TRAIN was never used to select the checkpoint; full TRAIN-fitted SBS.

Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
ALL	71.2403	66.0332	11.389208	11.400702	11.345073	-0.1973	+0.3875	0.3960
TSP	45.9580	32.8617	8.004212	8.023959	7.970215	-0.2461	+0.4265	0.4206
CVRP	47.4914	18.1284	18.039929	18.151987	17.946816	-0.6173	+0.5188	0.6853
ATSP	76.4642	46.9210	1.640048	1.678679	1.633938	-2.3013	+0.3739	0.3744
OVRP	41.1753	34.2420	7.443859	7.453531	7.397676	-0.1298	+0.6243	0.6369
VRPB	80.6185	80.2765	9.355575	9.357264	9.319342	-0.0180	+0.3888	0.3966
VRPL	37.7481	31.5309	12.393262	12.400882	12.344503	-0.0614	+0.3950	0.4024
VRPTW	80.3568	80.2407	18.812887	18.813703	18.757954	-0.0043	+0.2928	0.3064
OVRPTW	83.8753	83.8025	10.118134	10.118640	10.093220	-0.0050	+0.2468	0.2422
OVRPB	88.4099	88.3580	6.363062	6.363230	6.346658	-0.0026	+0.2585	0.2662
OVRPL	41.5049	34.4716	7.444883	7.453219	7.398483	-0.1118	+0.6272	0.6377
VRPBL	79.8605	79.4593	9.401835	9.403115	9.363196	-0.0136	+0.4127	0.4141
VRPBTW	79.6160	79.4914	19.731633	19.733549	19.642712	-0.0097	+0.4527	0.4375
VRPLTW	80.4877	80.3827	18.768766	18.769959	18.714366	-0.0064	+0.2907	0.3052
OVRPBL	88.5852	88.5556	6.358798	6.358857	6.342513	-0.0009	+0.2568	0.2643
OVRPBTW	83.2901	83.2432	10.623858	10.624446	10.586225	-0.0055	+0.3555	0.3298
OVRPLTW	83.7926	83.7321	10.110798	10.111265	10.085768	-0.0046	+0.2482	0.2444
VRPBLTW	79.7000	79.5630	19.769263	19.770814	19.680364	-0.0078	+0.4517	0.4337
OVRPBLTW	83.3914	83.3370	10.624946	10.625544	10.587365	-0.0056	+0.3550	0.3309

ALL task-equal; ratios of mean costs differ from per-instance actual regret.

R107/A_edge_factorized | Fixed target36 vs R89A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	154	73	150	0.000095688
VAL	4625	9	4	8	0.000047035
TEST	4598	6	1	4	0.000091091

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.004184	10353	10414	0.000095688	0.001322
VAL	0.006173	574	569	0.000047035	0.001019
TEST	0.028395	593	570	0.000091091	0.000440

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R107/A_edge_factorized | Fixed target36 vs R89B

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	138	750	182	0.001132015
VAL	4625	9	41	7	0.000585015
TEST	4598	7	31	8	0.000406125

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.450823	21994	28567	0.001132015	0.012389
VAL	-0.191358	1323	1478	0.000585015	0.006024
TEST	-0.159259	1315	1444	0.000406125	0.005029

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R107/A_edge_factorized | Fixed target36 vs R84A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	182	67	177	0.000048031
VAL	4625	11	2	8	0.000267821
TEST	4598	8	0	5	0.000033533

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.003567	12013	12065	0.000048031	0.000824
VAL	-0.028395	670	693	0.000267821	0.003206
TEST	0.024691	672	652	0.000033533	-0.000171

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.